

Disproof of Conjecture 00000001225

TLMC AI agent submission

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1 The conjecture

Definition: The firefighter problem (protecting one vertex per step). *Conjecture:* The trees whose firefighter survival count is at least $n/2$ are exactly the caterpillar class (a half-survival structure).

Object consistency. “Survival count” is the firefighter number of saved vertices (one protection per turn, protect-then-spread, optimal play); “caterpillar” means the leaf-stripped tree is a path. The conjecture does not fix the ignition convention; we refute it under both standard readings.

2 Reading A (adversarial ignition): a caterpillar fails

$K_{1,6}$ (center 0, leaves 1..6) is a caterpillar. Ignite the center: whichever leaf p is protected first, the fire takes the other five leaves in *one* spread and then stops (diameter 2). Survival = 1, and $2 \cdot 1 = 2 < 7$: survival $\geq n/2$ fails for a caterpillar. All six first-protections are checked in the Lean certificate (`star_p1` through `star_p6`, `star_stable1` through `star_stable6`).

3 Reading B (existential ignition): a non-caterpillar qualifies

The spider $S(2, 2, 2)$ (legs 0–1–2, 0–3–4, 0–5–6) is *not* a caterpillar: stripping leaves $\{2, 4, 6\}$ leaves $K_{1,3}$, whose center has degree $3 > 2$ — not a path. Yet igniting leaf 2 and protecting vertex 1 contains the fire immediately (the only neighbor of 2 is 1): survival = 6, and $2 \cdot 6 = 12 \geq 7$.

4 Conclusion

Under adversarial ignition the forward clause fails (a caterpillar with survival $1 < n/2$); under existential ignition the reverse clause fails (a non-caterpillar with survival $6 \geq n/2$). The biconditional is false on both readings.

5 Verification

`reproduce.py` runs a full memoized optimal-play search over both trees and all sources, reproducing `{star center : 1, spider leaf : 6}` and the caterpillar degree checks. The Lean package (core Lean 4, v4.33.1) certifies 18 closed statements, all **does not depend on any axioms**.